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Assessing urban flood resilience: a comprehensive framework with evidence from Melbourne

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Abstract

Flooding in urban areas poses significant challenges due to rapid urbanization, population growth, and changing climate patterns. This study introduces a comprehensive flood resilience assessment framework tailored for the local government areas (LGAs) of Moonee Valley and Maribyrnong in Melbourne. By integrating hydrological modeling, spatial analysis, and socio-economic data, we developed a Flood Resilience Index (FRI) and an Integrated Flood Resilience Index (IFRI) to evaluate flood exposure, sensitivity, and adaptive capacity across 474 urban units. Additionally, a Damage Index was introduced to estimate potential flood impacts, including property damage, infrastructure disruption, and economic losses, allowing a more detailed understanding of vulnerability and recovery capacity. Our findings reveal distinct spatial patterns in flood resilience, with higher exposure and sensitivity in certain areas due to socio-economic and environmental factors. Multi-scale Geographically Weighted Regression (MGWR) was applied to examine the spatially varying relationships between flood resilience and its contributing factors, revealing significant local heterogeneity and providing important insights for targeted policy interventions. Despite limitations related to data scope and future scenario representation, this framework offers a robust tool for urban planners and policymakers to enhance flood resilience, optimize resource allocation, and implement effective mitigation strategies. Future research should incorporate dynamic data and community engagement to further refine resilience assessments and support sustainable urban development.

Keywords Flood resilience, Damage index, IFRI, Exposure, Sensitivity, Adaptive capacity, MGWR, Analytical Hierarchy Process (AHP)

1 Introduction

1.1 Background

Flooding has become an increasingly critical concern in urban environments due to rapid urbanization, population growth, and evolving climatic conditions. Urbanization drives extensive land-use changes, replacing natural

landscapes with impervious surfaces that accelerate surface runoff and elevate flood risk (Over et al., 2025; Tang et al., 2025). Population growth further intensifies flood exposure by expanding human settlements into flood-prone areas, while the impacts of climate change, such as rising sea levels, intensified rainfall, and prolonged high-tide events, amplify hydrological hazards (Hsiao et al., 2024). Together, these factors exacerbate the frequency, magnitude, and socio-economic impacts of urban flooding, posing complex challenges for urban planning, infrastructure management, and community resilience. Recent research has increasingly focused on intense

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rainfall events, which involve large volumes of precipitation over a short period (Yamagishi et al., 2013). Such events can trigger rapid-onset floods, causing substantial property damage and posing serious risks to human life (Zhang et al., 2023). These challenges highlight the need for quantitative approaches that can accurately assess, monitor, and enhance urban resilience to intense rainfall and associated flooding.

Flooding is a recurrent feature in Greater Melbourne Area, Australia, costing an estimated \$735 million per year (Melbourne Water, 2023). According to the Victorian Floodplain Management Strategy (*Victorian Floodplain Management Strategy*, 2016), 59 flood studies have been undertaken at local and regional levels to make Victorians and local councils aware of the flood risks and provide measures to fight against the consequences of floods (Turner & Sun, 2024). In response to severe flooding in May 1974 along the Maribyrnong River and Merri Creek, a telemetry-based flood warning system was established for Melbourne's major waterways in 1975 (Giesemann & Crapper, 1987). The system consists of sixty-seven stations, each sending rainfall and/or stream level data every six minutes to a central computer located in the Melbourne and Metropolitan Board of Works head office. Officers can predict the expected flood magnitudes, arrival times, and likely flooded areas and this information is shared with the relevant authorities, enabling appropriate actions to be taken.

Communities often don't prioritize risk reduction and vulnerability measures until a disaster strikes, even if they are aware of the potential hazards in their area (Xu et al., 2020). Vulnerability is determined by three factors: exposure (identifying who or what is at risk), sensitivity (measuring how susceptible a system is to harm), and coping mechanisms (strategies to deal with hazards). Resilience, in contrast, refers to a system's ability to absorb disturbances and reorganize effectively, showcasing traits such as reliability, adaptability, speed, and robustness (Xu et al., 2020). Resilience can be demonstrated in actions such as absorbing damage, adjusting to risks, preventing disasters from worsening during recovery, and minimizing their overall impact (Nelson et al., 2007; Thomas, 2022). Community resilience can reduce damages through human intervention, but its effect is limited in situations where interventions are less effective (Gawith et al., 2016). It primarily helps mitigate impacts such as property damage, economic losses, health risks, and displacement.

Assessing flood vulnerability and enhancing flood resilience is crucial in mitigating the impact of flood disasters in affected regions (Ali et al., 2020). The resilience of communities in managing flood impacts is influenced by a combination of factors, including people's exposure and their sociodemographic and socioeconomic traits,

such as employment, education, household composition, and demographic structure, as well as the community's capacity to manage hazards and recover from them (Díez-Herrero & Garrote, 2020). Strengthening flood resilience by combining community preparedness, early warning systems, and infrastructure improvements can effectively reduce flood damage, especially as climate change intensifies (Miyamoto et al., 2022). Investing in improving resilience at both the national and community levels is a key strategy to minimize the impact of disasters on the country and its local areas and enhancing resilience to disasters is a collective effort involving citizens, businesses, and the government (National Academies et al., 2012). Also, understanding the influencing mechanisms of urban flood resilience is essential for enhancing long-term urban flood resilience in the face of global climate change and urbanization (Zhu et al., 2023). Despite the abundance of technical literature focusing on flood hazard analysis, vulnerability, and resilience, there remains a lack of comprehensive scientific studies and practical expertise in conducting integrated flood risk assessments (Díez-Herrero & Garrote, 2020). The existing research resources, such as hazard data and maps, do not effectively aid in prioritizing mitigation measures, even though risk mapping does help in identifying at-risk communities.

Various frameworks have been derived globally to aid flood risk management, each serving specific purposes. Depth-damage functions are widely used in flood damage assessment that provides a mathematical relationship between the depth of flooding and the expected damage to different types of structures and land use (Gerl et al., 2016). The Social Flood Vulnerability Index (SFVI) has been assessed for Moreland City in northern Melbourne exploring the social component of the Flood Vulnerability Index (FVI) and focusing on exposure, susceptibility, and resilience factors (Rashetnia et al., 2015). Most existing studies on resilience assessment remain theoretical, lacking clear applications for policymaking and practical resilience efforts (Zhang et al., 2023). Challenges include the difficulty in quantifying indexes and variations in subjective assessments across regions, making comparisons complex. Some studies highlighted the importance of community engagement, interviews, surveys, training programs, improved government policies, and awareness. All these enhance the qualitative assessment of resilience but do not contribute to the quantitative assessment (Restemeyer et al., 2015; Rogers et al., 2020; Shah et al., 2018). Some studies indicated the limitations in their quantitative assessment of resilience due to the unavailability of quality and relevant data (Chen et al., 2022).

To overcome these issues, this study introduces a comprehensive quantitative framework for assessing flood

resilience, enabling regional comparisons and analysis of resilience trends. Based on the framework, potential local social, policy, technical, and design interventions are identified to enhance flood resilience. Where implementing adaptation measures is not feasible, the framework guides the optimal siting of a new Emergency Management Centre based on minimum travel time to rescue points.

The impact of floods on urban systems is commonly assessed through flood-induced damage, which includes both direct damage, such as physical destruction of infrastructure and property, and indirect damage, such as economic losses, disruption of production, healthcare expenses, and impacts on human well-being (Cochrane, 2004; Zhang et al., 2023). Direct damages capture the immediate effects of floodwaters on buildings, roads, and critical infrastructure, while indirect damages reflect the broader socioeconomic consequences, including reduced income, business interruptions, and long-term health impacts. To evaluate flood resilience in a structured way, this study integrates a Damage Index, which estimates potential impacts of floods on property, infrastructure, and economic activity. By combining this with traditional resilience metrics: exposure, sensitivity, and adaptive capacity, we provide a more comprehensive understanding of the ability of communities to withstand and recover from flood events. The relationship between disaster intensity and damage is nonlinear, increasing rapidly past a threshold (Zhang et al., 2023). Higher damage indicates lower resilience, while higher resilience enables the system to endure more intense disasters with less damage.

This study introduces two complementary indices: the FRI and the IFRI. The FRI measures the inherent resilience potential based on exposure, sensitivity, and adaptive capacity indicators. The IFRI extends this by integrating observed flood damage outcomes, linking theoretical resilience capacity with observed flood impacts. Thus, while the FRI reflects potential resilience, the IFRI assesses effective resilience, enabling comparison between capacity and empirical damage outcomes. Building on this framework, the study has three main objectives, each linked to a guiding research question:

1. Develop a spatially informed framework for assessing urban flood resilience by integrating hydrological, environmental, and socio-economic indicators.
2. RQ1: What are the key components required to construct an effective flood resilience assessment framework for urban areas?
3. Construct and apply the FRI and IFRI to evaluate spatial variations in flood resilience across urban units.

4. RQ2: How do flood exposure, sensitivity, and adaptive capacity vary spatially across the Moonee Valley and Maribyrnong LGAs?
5. Analyse spatial patterns of resilience to identify vulnerable areas and inform planning and policy interventions.
6. RQ3: What spatial patterns of flood resilience and vulnerability emerge, and how can these insights support targeted planning and risk reduction?

1.2 Literature review

1.2.1 Social vulnerability and flood resilience

Flood resilience research increasingly recognizes that social vulnerability, shaped by socioeconomic, demographic, and institutional factors, plays a critical role in determining a community's capacity to prepare for, respond to, and recover from flood (Rufat et al., 2015). Studies such as Tascón-González et al. (2020) and Englund et al. (2023) developed social vulnerability indices emphasizing demographic and economic inequalities, education, and institutional preparedness as key drivers influencing flood impacts. Similarly, Stone and GISP and Liu et al. (2021) highlighted that urban flood risk is not solely determined by physical exposure but also by uneven social capacities, especially among low-income or health-compromised populations.

In China, Yang et al. (2021) applied data envelopment analysis to reveal that regions with low vulnerability may still exhibit weak resilience, stressing the non-linear relationship between exposure and adaptive capacity. Shah et al. (2018) and Khatooni et al. (2023) reinforced the importance of social and institutional factors, such as community awareness, employment diversity, and governance, in mitigating vulnerabilities. However, despite diverse approaches, few studies integrate these indicators within a spatially explicit resilience framework at the urban or local government level.

1.2.2 Flood exposure and risk assessment

Flood exposure mapping has been widely explored through geospatial and multi-criteria analysis techniques (Paprotny et al., 2019). Rogers et al. (2020) emphasized a holistic approach to urban flood resilience in Melbourne, integrating community engagement, urban design, and policy responses. Their findings highlight that combining modelling outcomes with participatory decision-making improves local adaptation strategies an insight that guides the current study's integration of social and spatial dimensions. Meanwhile, Chen et al. (2022) advanced multi-criteria flood resilience evaluation through the VIKOR-GRA method, incorporating technical, organizational, social, and economic (TOSE) factors. Although comprehensive, their framework lacked a detailed

physical exposure component, which this study addresses through integration with hydrological modelling.

1.2.3 Hydrodynamic modelling and HEC-RAS applications

Hydrodynamic models such as HEC-RAS are essential tools for simulating flood inundation and depth (Urzică et al., 2020). Applications of HEC-RAS in flood risk assessment have demonstrated its accuracy and flexibility across varied topographies (Abedin & MacVicar, 2024; Sampson et al., 2015). For instance, Liu et al. (2021) applied HEC-RAS in southern Taiwan to assess urban flood risks, integrating hydrological outputs with socio-economic indicators to identify priority areas for intervention. Similarly, Abdel-Mooty et al. (2021) introduced data-driven methods linking physical simulations with resilience indicators such as robustness and recovery time. However, most studies stop short of coupling these models with social and adaptive capacity indices. The current study builds on this foundation by integrating HEC-RAS-derived flood parameters with socio-economic and demographic data, yielding a more comprehensive flood resilience index.

1.2.4 Integrative frameworks and identified gaps

Recent interdisciplinary approaches, such as Restemeyer et al. (2015) and Sen et al. (2021), argue that effective flood resilience assessment requires combining technical, social, and policy domains. However, these frameworks often rely heavily on qualitative assessments or expert judgment, limiting reproducibility and comparability across regions. Furthermore, while studies like Khatooni et al. (2023) attempted to integrate fuzzy logic and hydraulic simulations, many assigned equal weights to indicators without empirical justification.

Despite extensive literature on flood vulnerability and exposure, a notable gap remains in developing a quantitative, multi-dimensional resilience framework that unites physical (exposure), social (sensitivity), and adaptive (capacity) components at a local scale. This study addresses these limitations by developing a Comprehensive Flood Resilience Assessment Framework combining hydrodynamic modelling (HEC-RAS) with socio-economic indicators to evaluate flood resilience in Moonee Valley and Maribyrnong. The framework enhances existing approaches by incorporating weighted indicators derived from spatial analysis, allowing for the identification of local disparities and the formulation of evidence-based adaptation strategies. Although extensive literature exists on the physical and social dimensions of flood risk, most studies examine these aspects in isolation, resulting in fragmented resilience assessments. This research contributes to bridging that divide by developing a spatially explicit, data-driven framework that integrates exposure, sensitivity, and adaptive capacity into a unified resilience

assessment model. By combining HEC-RAS flood simulations with socioeconomic and demographic data, the framework offers a replicable and policy-relevant approach for assessing local-scale flood resilience, providing valuable insights for adaptive planning and equitable disaster management in urban environments.

2 Methodology

2.1 Study area and datasets

The study focuses on Maribyrnong and Moonee Valley LGAs in Great Melbourne Area, Victoria State, Australia. Moonee Valley (Fig. 1), located to the north-west of Melbourne is positioned between Melbourne CBD and the Melbourne airport covering 43KM² and a population of 129,379 in 2022 (*Priority Projects: Community Vision, Council Plan and Health Plan Implementation Update Quarter One 2022/23: July to September*, 2023) while Maribyrnong (144°50'17" -144°54'59"E, 37°45'21"-37°49'37"S), is situated south of Moonee Valley which is 5 to 11 km west of the Melbourne CBD covering area of 31.21 square kilometres and population of 87,355 (as of June 2016) (*City snapshot*, 2023). The terrain is predominantly flat, with low-lying floodplain areas along river corridors and gradually rising urban areas inland. Both LGAs contain Maribyrnong River as a major waterway which serves as a significant focal point for flooding in both areas during periods of heavy rainfall or extreme weather conditions. Other than Maribyrnong River, the city of Maribyrnong comprises Stoney Creek as another prominent waterway and the city of Moonee Valley comprises Moonee Ponds Creek and Steele Creek as the major stream systems (*Storm and Flood Emergency Plan*, 2018; *Storm and Flood Emergency Plan*, 2021). Both municipalities can be affected by rising levels of the rivers and creeks (riverine flooding) or heavy rainfall over a brief period (flash flooding). Storm and Flood Emergency Plan (2018 and 2021) indicated that the high tides incoming from Port Philip Bay and adding to heavy rainfall worsen the flooding in these municipalities. As the municipalities are close to Port Philip Bay and have flat terrain, tidal flows from the bay during extreme storm events can aggravate the flooding issues in these areas. The Maribyrnong River has experienced frequent overflow incidents, with more than 25 recorded occasions documented since 1871. The 1974 Maribyrnong River flood, ranking as the third largest in the region's history, caused significant damage to residential, industrial, and public structures, including Flemington Racecourse, summing approximately \$3–5 million in 1974 values which is equivalent to \$19–31 million today, with residential properties comprising \$0.6 million of the total cost (*Storm and Flood Emergency Plan*, 2021).

For this study, land use, Digital Elevation Model (DEM), rainfall and river data, and socio-economic

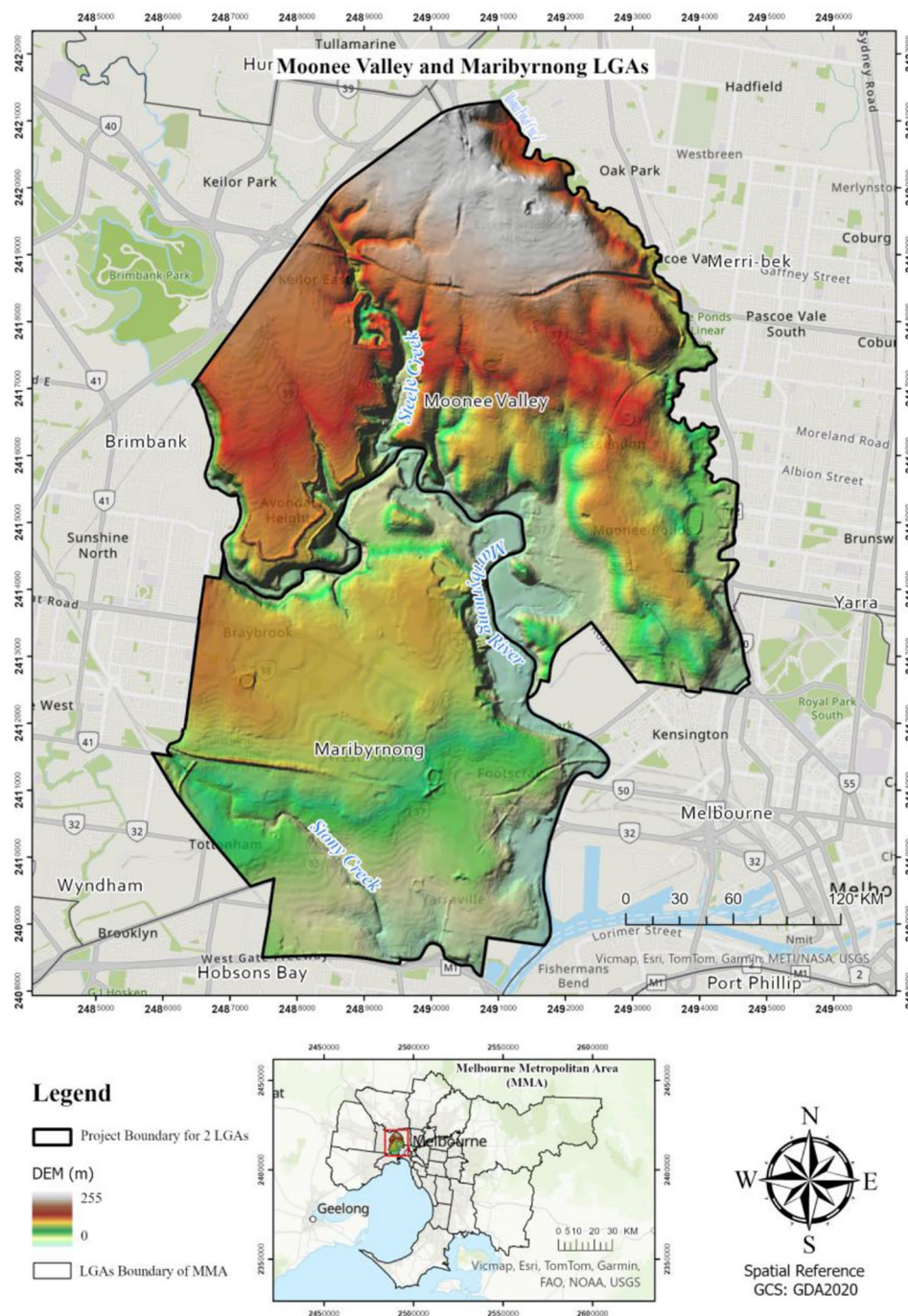


Fig. 1 Study area of Maribyrnong and Moonee Valley LGAs

variables, such as dwellings, total and dependent population, population with long-term health conditions, employment, education, and average income, were used for both Moonee Valley and Maribyrnong LGAs. The study analyzed 474 Statistical Areas Level 1 (SA1s), referred to here as “urban units,” within the Maribyrnong and Moonee Valley LGAs, Victoria, Australia. SA1s are the smallest spatial units defined by the Australian

Bureau of Statistics for census data, typically containing 200–800 residents (average ~400) and varying in area depending on population density. Flooding simulation and resilience assessment were conducted using a scenario-based hydrological model developed from LiDAR DEM (10 m resolution), rainfall, and river datasets within a GIS environment, and implemented using the Hydrologic Engineering Centre’s River Analysis

System (HEC-RAS). Land use data was used to determine the potential damage to different land use types of Moonee Valley and Maribyrnong LGAs. Similarly, socio-economic parameters were used to understand the socio-economic impact of flooding for both municipalities. The dependent population includes individuals aged below 20 years and over 59 years. The population with long-term health conditions comprises individuals diagnosed with at least one chronic illness, such as cancer, asthma, or similar conditions. Also, the mean income of these two LGAs is \$91,000-\$103,999 per annum. Mean income-generating population data is based on this range of mean income.

The tabulated form of the datasets with their source of information, data format, and their intended use during the research is shown in Table 1:

2.2 Resilience assessment

The proposed resilience assessment framework (Fig. 2) for urban flooding involves a detailed process to evaluate flood resilience. Here's a breakdown of each step:

2.2.1 Data collection and preprocessing

This phase involved gathering relevant datasets such as DEMs, precipitation data, river flow rates, sand percentage in soil, land use, and all socioeconomic indicators like population, number of dwellings, population with long-term health conditions, dependent population aged less than 20 years and more than 59 years, employed

population, educated population and mean income generating population for all statistical areas of level 1 of Maribyrnong and Moonee Valley LGAs of Victoria.

Data preprocessing tasks included georeferencing, data cleaning, resolution matching, and coordinate system alignment to ensure consistency and compatibility among different datasets.

2.2.2 Flood depth generation

Flood prediction and flood management comprises the collection of comprehensive geospatial data, operation of statistical spatial analysis, and implementation of hydrological and hydrodynamic models (Pinho et al., 2015). LiDAR-derived topographic data like DEM, and hydrologic data like precipitation, and river flow discharges were collected. All these data were imported into modeling software, HEC-RAS to build the hydrological models and simulate the model to analyze the water flows, inundation areas, and their interactions with the terrain. Using hydraulic modeling software HEC-RAS, flood depths were simulated based on input data such as DEMs, precipitation, and river flow rates. HEC-RAS computed flood extents and depths for different scenarios, providing spatially distributed flood depth information for the study area. Observed floodplain maps reflect past events and limited extents, whereas HEC-RAS allows scenario-based 2D flood simulations, capturing spatial variability under different rainfall and river flow conditions. HEC-RAS outputs were validated against Maribyrnong River

Table 1 Datasets with their data types, authority, and their justification for use

Description	Format	Authority	Year/ Resolution	Justification for Use
Project Boundary	Esri Shapefile	Australian Bureau of Statistics (ABS)	2022	To set the boundaries of the project.
Statistical Area Level 1 (SA1s) Boundary and Statistical Area Level 2 (SA2s) Boundary	Esri Shapefile	Australian Bureau of Statistics (ABS)	2021	To delineate the areas for detailed spatial analysis.
Vicmap Elevation DEM 10 m	Geo TIFF	Victorian State Government Department of Environment, Land, Water and Planning	2021/10m	To simulate urban flooding from the developed hydrological model, identify low-lying areas prone to flooding, and model flood scenarios
Land use	Geo TIFF	Esri, ArcGIS Living Atlas of the World	2023/10m	To identify which types of land use are more susceptible to flooding
MMA Administrative Boundary	Esri Shapefile	Australian Bureau of Statistics (ABS)	2022	To delineate the administrative boundary of MMA
Socio-economic data (Population, Number of dwellings, Dependent population (age group < 20 and > 59), Population with Long Term Health Conditions, Employed Population, Educated Population, Mean Income Generating Population)	Excel file	Table Builder, Australian Bureau of Statistics (ABS)	2021/SA1	To assess potential impacts and importance of socio-economic data on flood resilience and vulnerability.
Soil Grids of Victorian- Sand Percentage	Geo TIFF	Victorian State Government Department of Environment, Land, Water and Planning	2021/30m	To assess the sand composition in soil and its influence on flood resilience.
Rainfall and River Flow Rate Data	Excel file	Melbourne Water	2022/hourly	To run flood simulation and prediction processes.

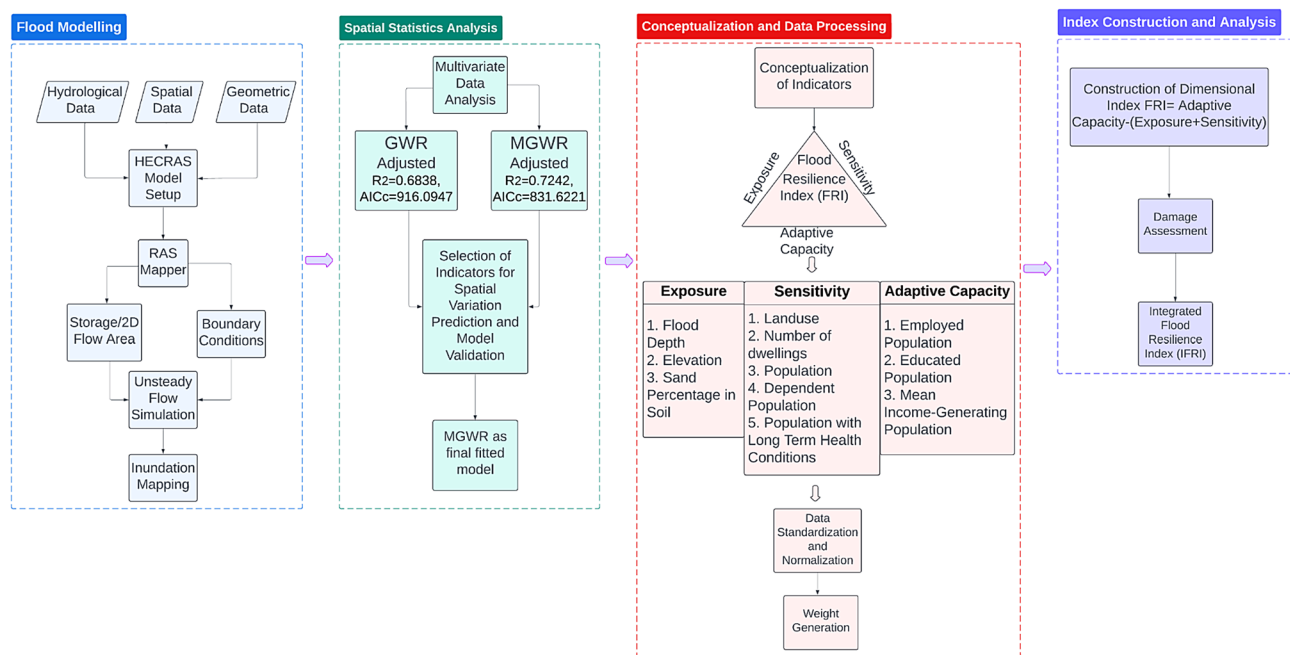


Fig. 2 Methodological framework

flood extents (Storm and Flood Emergency Plan, 2018), showing a high level of agreement ($\pm 10\%$ spatial deviation). This approach enables predictive resilience assessment rather than only historical mapping.

2.2.2.1 HEC-RAS modelling For a two-dimensional (2D) analysis, the projection data was first provided to the River Analysis System (RAS) mapper. The RAS Mapper tool facilitated the direct import of necessary terrain data into the model. A storage area shape file from ArcGIS was imported into the geometry, enabling the acquisition of volume-elevation relationships from the loaded terrain data. This volume-elevation information was directly applied to the storage area. A 2D flow area was created with a grid size of 50 m by 50 m, using a total of 7791 cells for the analysis. Additionally, default Manning's n value 0.06 representing land cover roughness, was imported into the HEC-RAS geometry editor interface. It is important to note that for accurate characterization of unstable flow events over the simulation period, the model cannot incorporate roughness changes caused by flood waves (Karki et al., 2022). Setting up the external boundary conditions was the first step in using HEC-RAS to model the downstream channel. The upstream boundary was selected to be independent of the conditions further downstream. The downstream boundary was chosen to be independent of the flow characteristics beyond it. All scenarios in the HEC-RAS model required the same flow and boundary conditions (Abedin & MacVicar, 2024; Sampson et al., 2015). Additionally, precipitation was included as a boundary condition within the 2D flow area.

At the downstream end of the study area, a normal depth boundary condition was applied. The simulation ran from October 14, 2022, at 0:00 to October 15, 2022, at 23:00. This period was chosen because it coincides with the most recent historical flood event on the Maribyrnong River. Rainfall data were downloaded from Melbourne Water's monitoring station within the Maribyrnong River catchment at a 1 h temporal resolution, while corresponding river flow data from the Maribyrnong River gauge were used as boundary conditions. A fixed time step of 1 min was used for computations. The output for mapping was set to be recorded every 10 min, while the hydrograph and detailed outputs were recorded every hour.

2.2.3 MGWR analysis

Multi-scale Geographically Weighted Regression (MGWR) extends traditional Geographically Weighted Regression (GWR) by allowing explanatory variables to operate at different spatial scales, thereby capturing spatial heterogeneity in relationships.

In this study, MGWR is applied to explore the spatially varying relationships between flood-related outcomes and their associated environmental and socio-economic drivers. The model estimates location-specific regression coefficients, enabling the identification of how the influence of different factors changes across space.

Flood depth is used as the dependent variable, while elevation, sand percentage in soil, land use, and socio-economic indicators, including population, number of dwellings, population with long-term health conditions, dependent population (aged under 20 and over 59),

employed population, educated population, and mean income-generating population, are included as explanatory variables. This formulation allows the analysis to capture local spatial variations in flood conditions and their associations with physical and socio-economic characteristics.

2.2.4 Conceptualization of indicators

Indicators are categorized into three dimensions of flood vulnerability: exposure, sensitivity, and adaptive capacity.

Exposure indicators reflect the physical characteristics of the study area susceptible to flood hazards. These indicators provide insight into the degree to which a specific region is likely to experience flooding (Balica et al., 2009; Cutter et al., 2010). For example, the following indicators- flood depth, elevation, and sand percentage in soil are categorized in exposure dimension. Flood depth measures the height of water during a flood event, indicating the severity of flooding in various areas. Regions with greater flood depths are at higher risk of damage to infrastructure and property. The altitude of the land above sea level influences flood exposure. Lower elevations are generally more prone to flooding, especially in coastal and riverine areas. The composition of the soil, particularly the amount of sand, affects water absorption and runoff. Soils with higher sand content tend to have better drainage but can also lead to rapid water flow and erosion during floods. Although urban areas are largely impervious, the sand percentage remains relevant in semi-urban, green, or peri-urban zones within the study area, where natural or exposed surfaces still affect floodwater retention and flow dynamics. This indicator therefore captures residual hydrological variability that contributes to flood exposure across urban units.

Sensitivity indicators represent socio-economic factors that influence the community's vulnerability to floods. These indicators highlight the community's inherent vulnerabilities due to their social and economic characteristics (Tascón-González et al., 2020). For example, the following indicators- land use, number of dwellings, population, dependent population, and population with long-term health conditions are categorized in the sensitivity dimension. The way land is utilized, such as residential, agricultural, or commercial purposes, affects flood vulnerability. For instance, urban areas with impermeable surfaces are more prone to severe flooding. The total number of residential structures in an area can indicate potential housing losses and the number of households affected by flooding. The overall population size provides a measure of the number of individuals potentially at risk. Dependent Population includes children and, the elderly, who are less able to respond to and recover from flood events on their own. Individuals with chronic illnesses

are more vulnerable to the health impacts of flooding, including disruption of medical care and increased risk of waterborne diseases.

Adaptive capacity indicators measure the ability of the community to respond and recover from flood events. These indicators reflect the resources and capabilities available to reduce flood vulnerability (Rufat et al., 2015; Smit & Wandel, 2006). For example, the following indicators- employed population, educated population, and mean income-generating population are categorized in the adaptive capacity dimension who can contribute to the economic stability and recovery efforts of the community. Higher educational qualifications enhance individuals' ability to secure better employment opportunities, access information, and make informed decisions during and after flood events. Higher educational qualifications are the most practical way to address the resilience of an area (Majumder et al., 2023). This improves overall community resilience. The average income level of the population indicates the financial resources available for recovery and adaptation measures. Higher-income levels correlate with greater access to resources and services necessary for effective flood management. Table 2 lists indicators relevant to the study along with their weighting on how they influence the respective risk component.

2.2.5 Weighting criteria calculation

All indicators were first standardized to a mean of zero and standard deviation of one:

$$Z = \frac{(X - \mu)}{\sigma}$$

They were then normalized to a 0–1 range:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

The Analytical Hierarchy Process (AHP) was applied to determine the relative importance (weights) of each indicator within exposure, sensitivity, and adaptive capacity dimensions.

The AHP, developed by Saaty (Saaty, 1977), is a multi-objective, multi-criteria decision analysis method that employs a pair-wise comparison process to prioritize choices among a set of alternatives (Fernández & Lutz, 2010). AHP uses a basic scale of absolute numbers (as shown in Table 3) to represent individual preferences (Saaty & Vargas, 2007). The AHP method is applied to determine the relative importance of indicators within each vulnerability dimension. Pairwise comparisons are used to assign weights to indicators based on their perceived significance in contributing to flood resilience. A

Table 2 Identified dimensions of flood resilience, corresponding indicators, their relevance and assigned weight

Components	Indicators	Justifications	Criteria Weights
Exposure	Flood Depth	Flood depth is a direct measure of the severity of flooding events. Deeper floods can cause more significant damage to infrastructure, property, and pose higher risks to human life. This indicator is crucial for understanding the potential impact of flood events on urban areas.	0.019
	Elevation	Elevation affects flood susceptibility as lower areas are more likely to experience flooding. Areas at higher elevations are generally less vulnerable to floodwaters, making elevation an essential factor in assessing flood risk.	0.014
	Sand Percentage in Soil	The sand percentage in soil influences soil permeability and drainage capacity. Soils with higher sand content typically have better drainage, reducing the likelihood of waterlogging and flood accumulation. This indicator helps evaluate the natural ability of the land to cope with heavy rainfall and flooding.	0.014
Sensitivity	Land use	Land use influences the extent of impervious surfaces, affecting runoff and floodwater absorption. Urban areas with more impervious surfaces are more prone to flooding, making land use a key factor in flood sensitivity.	0.050
	Number of Dwellings	The number of dwellings reflects population density and housing concentration. Areas with more dwellings are likely to experience higher damage and displacement during floods, making this an important sensitivity indicator.	0.050
	Population	Population size is linked to the potential impact on human life and the need for evacuation and emergency services. Larger populations require more resources for disaster response and recovery, increasing flood sensitivity.	0.074
	Dependent Population	The dependent population (age groups < 20 and > 59) includes individuals who are often more vulnerable during disasters. These age groups may have limited mobility and higher health care needs, making them more sensitive to flood impacts.	0.074
	Population with Long Term Health Conditions	Individuals with long-term health conditions are at higher risk during floods due to their medical needs and potential mobility issues. This group requires additional support and resources, increasing the overall sensitivity of the population to flooding events.	0.110
Adaptive Capacity	Employed Population	Employment indicates economic stability and recovery potential. Employed individuals have financial resources for repairs, rebuilding, and accessing emergency services, enhancing adaptive capacity.	0.168
	Educated Population	Education correlates with disaster awareness and preparedness. An educated population is more likely to know about flood risks and mitigation strategies, improving the community's ability to respond and adapt.	0.168
	Mean Income	Higher income levels indicate economic resilience. Greater financial resources allow for investment in flood prevention, insurance, and recovery efforts, making this a critical indicator for assessing financial capacity to withstand floods.	0.259
	Generating Population		

Table 3 Saaty's scale of preference

Numerical Values	Description
1	Equal importance of both elements
3	Moderate importance of one element over another
5	Strong importance of one element over another
7	Very importance of one element over another
9	Extreme importance of one element over another
2, 4, 6, 8	Intermediate values between two adjacent values

scale is used to assign numerical values representing the importance or preference of one criterion over another. The scale often ranges from 1 to 9, with 1 representing equal importance and 9 representing extreme importance. A key feature of AHP is its ability to assess the consistency of these preferences using a numerical process. After completing pairwise comparisons, a consistency check is performed to ensure the comparisons are logical and consistent. This is achieved through the consistency ratio (*CR*), which indicates the reliability of the judgments made. For the preferences to be considered

consistent, the *CR* should be less than or equal to 0.10. The consistency ratio is defined as:

$$CR = \frac{CI}{RI}$$

where:

CI is the consistency index, and.

RI is the random index.

The value of *RI* depends on the number of criteria (*n*).

The consistency index (*CI*) is calculated as:

$$CI = \frac{(\lambda_{max} - n)}{(n - 1)}$$

Here,

λ_{max} is the largest eigenvalue of the comparison matrix, and *n* represents the number of criteria (in this case, *n* = 11).

For this study, the *CR* was calculated to be 0.042, and the *CI* was found to be 0.064. The *RI* for *n* = 11 was

determined to be 1.51 for $n = 11$ (Table 4). The indicators were then assigned their respective weights (W) based on the consistent comparison matrix.

2.2.6 Calculation of Flood Resilience Index (FRI)

The weighted sum of normalized indicators within each dimension is computed by multiplication of normalized values of indicators with their respective weights to obtain composite scores for exposure, sensitivity, and adaptive capacity. These composite scores provide insights into the relative vulnerability of different regions within the study area. The general equation for calculating the dimension of the resilience index is given as follows:

$$Dimension = \sum_{n=1}^n (w_n * norm\ i_n)$$

where:

Dimension represents the overall score for a particular dimension of the resilience index (exposure, sensitivity, or adaptive capacity),

w_n is the weight assigned to the n -th indicator,

$norm\ i_n$ is the normalized value of the n -th indicator,

n is the total number of indicators considered for this dimension.

The FRI is then derived by combining the adaptive capacity score with the inverse sum of sensitivity and exposure scores. It is calculated by using the formula:

$$FRI = Adaptive\ Capacity - (Sensitivity + Exposure)$$

where, *FRI* quantifies the overall resilience of each location, considering both its ability to withstand flood impacts and its capacity to recover and adapt.

2.2.7 Damage assessment

The initial calculation of flood resilience in the study area was based on spatial variations and did not quantify resilience relative to the damage each area sustained. To address this, the damage to various socio-economic aspects caused by flood depth was calculated. This involved determining the ratio of the flood depth area to the total area of each SA1 and multiplying it by the socio-economic values of that SA1, assuming each indicator is proportionally distributed in each statistical area. It is important to note that this approach assumes a uniform distribution of the population and that structures are at ground level across each SA1. The socio-economic indicators considered were Land use, Number of Dwellings, Population, Dependent Population, Population with Long-Term Health Conditions, Employed Population, Educated Population, and Mean Income Generating

Table 4 Random Index values for the number of indicators used

Random Index (RI)		2	3	4	5	6	7	8	9	10	11	12	13	14	15
<i>n</i>	1	0.00	0.00	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.48	1.56	1.57	1.58

Source: (Yap et al., 2017)

Population. The damage for each indicator was standardized and normalized. These values were then summed to measure the total damage quantity for Maribyrnong and Moonee Valley LGAs. After calculating the total damage, it was standardized and normalized to create a Damage Index.

For example, damage for the population of a statistical area is calculated by:

$$D_{population} = \left(\frac{A_1}{A} \right) * P_1$$

where:

$D_{population}$ is damage on population,

A_1 is the area covered by flood depth,

A is the total area of a statistical boundary,

P_1 is the population of that statistical boundary.

The damage index is further utilized to measure the integrated flood resilience index based on the damage.

2.2.8 Integration of flood resilience and damage

This Damage Index was then integrated with the previously calculated FRI to produce the final IFRI. The formula used is:

$$IFRI = w_1 * FRI - w_2 * Damage\ Index$$

In this formula, w_1 and w_2 represent the weights assigned to each component. In this study, both components were given equal importance, so w_1 and w_2 were both set to 0.5. Assigning equal weights ($w_1 = w_2 = 0.5$) ensures that both dimensions contribute equally to the composite index, avoiding bias toward either physical or social factors in the absence of empirical evidence suggesting one is more influential than the other. This approach is also consistent with prior flood resilience studies and index-based assessments (e.g., Cutter et al., 2010) where equal weighting was applied when combining different but equally relevant dimensions of resilience.

IFRI provides a comprehensive measure of flood resilience that accounts for both spatial variations in vulnerability and the actual damage experienced by the community. Derivation of positive IFRI indicates that the community's resilience measures are strong enough to exceed the total damage caused by floods. A higher positive value would imply greater resilience relative to the damage experienced. Negative IFRI indicates that the total damage exceeds the community's resilience capacity. A larger negative value would suggest greater vulnerability and a need for enhanced resilience measures.

3 Results and discussions

3.1 Results of flood simulation

The flood simulation ran from October 14, 2022, at midnight to October 15, 2022, at 11:00 PM. This period was selected to capture realistic hydrological conditions and peak flows for validation. Simulated flood extents and depths were compared with observed floodplain maps and documented flood extents from the Maribyrnong River Flood Emergency Plan (2018). The results showed a high level of agreement, with spatial deviations within $\pm 10\%$, confirming the reliability of the model outputs. The values calculated for flood depth in the hydraulic models were transferred from HEC-RAS to ArcGIS software. The RAS Mapper tool is utilized to visualize water flow in a specific area over a given period and at a chosen time interval. From the flood simulation, there is a noticeable overflow downstream of the Maribyrnong River. Maximum velocities range from 1.75 to 2.05 m/s, and the maximum water level fluctuates between 5.8 and 6.9 m. Maximum velocities during these floods were notably high in the smaller sections of the riverbed. However, in the larger sections, they generally remain below 1 m/s. The flow is torrential throughout the area. These findings are essential for risk assessment, planning mitigation measures, and identifying the most vulnerable areas for future similar events (Diani et al., 2024) (Fig. 3).

3.2 Determinants of spatial variation of IFRI

All indicators used in the computation of the Integrated Flood Resilience Index (IFRI), including flood depth, elevation, sand percentage in soil, land use, number of dwellings, total population, dependent population, population with long-term health conditions, employed population, educated population, and mean income-generating population, were selected as explanatory variables for the MGWR analysis. The purpose of this analysis is not to validate the IFRI as an independent outcome, but to examine the spatially varying relationships between flood resilience and its contributing factors across neighborhoods. MGWR was applied to explore local spatial heterogeneity and to identify how the influence of resilience-related indicators varies geographically within the study area. This approach provides place-based insights into the relative importance of environmental, demographic, and socio-economic factors affecting flood resilience, supporting more targeted planning and policy interventions at the neighborhood level.

The spatial regression model results are shown in Table 5. The GWR model explains 68.38% of the variability with an Adjusted R^2 of 0.6838 and an AICc of 916.0947. The MGWR model has a higher Adjusted R^2 of 0.7242 (5.9% better than GWR) and a lower AICc of 831.6221, indicating a stronger capacity to capture local spatial heterogeneity in the IFRI. These results suggest

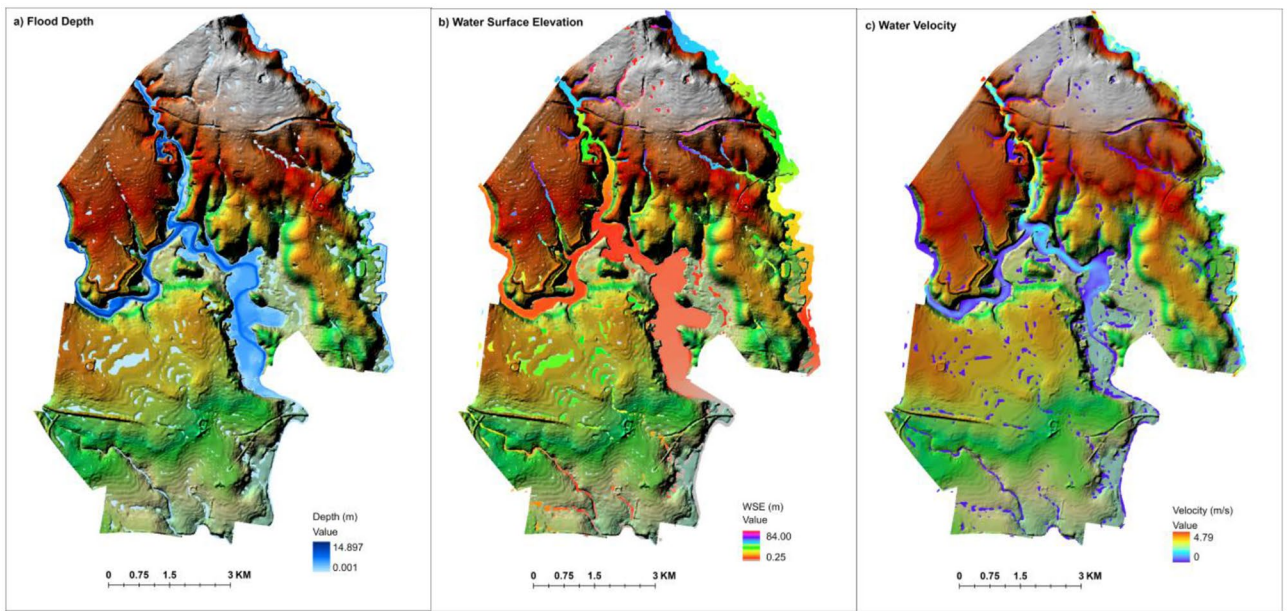


Fig. 3 Results of flood simulation (a) Flood Depth, (b) Water Surface Elevation, and (c) Water Velocity

Statistics	GWR	MGWR
R – Squared	0.7546	0.7669
Adjusted R-Squared	0.6838	0.7242
AICc	916.095	831.622
Sigma-Squared	0.316	0.2757
Sigma-Squared MLE	0.2454	0.2331
Effective Degree of Freedom	368.109	400.647
Optimal GWR Bandwidth: 62 (K nearest neighbours).		

that MGWR better represents neighborhood-scale variations in the relationships between resilience and its contributing factors. Table 5 provides the model specifications and diagnostic results for both GWR and MGWR models. The lower AICc and higher Adjusted R^2 of the MGWR model confirm its superior performance in explaining local spatial variation compared with the traditional GWR model.

Figure 4(a) shows the spatial distribution of local R^2 values from the MGWR model. High explanatory power (generally >0.610) is observed across most SA1s in the study area, particularly in the eastern and northern regions. Lower local R^2 values are concentrated in parts of the southern-central area, including SA1s in West Footscray–Tottenham and Braybrook, indicating weaker local model performance in these flood-prone zones.

Figures 4(b–k) illustrate the spatial distribution of MGWR coefficients for each explanatory variable. The results reveal clear spatial heterogeneity in how environmental and socio-economic factors influence flood resilience. Stronger coefficient effects are generally observed in central and western SA1s, while weaker or more

variable effects are found in eastern and flood-prone river-adjacent areas, highlighting distinct spatial patterns in resilience drivers across the study area.

3.3 Spatial variation of urban flood resilience dimensions

The study examines urban resilience of 474 SA1s in the Maribyrnong and Moonee Valley LGAs of Victoria. It assesses resilience across three main dimensions: exposure, sensitivity, and adaptability. Figures 5(a–d) illustrate the spatial distribution of flood exposure, sensitivity, adaptive capacity, and overall resilience across the 474 SA1s, revealing clear spatial heterogeneity driven by underlying environmental and socio-economic conditions.

3.3.1 Urban flood exposure dimension

Figure 5(a) presents the spatial distribution of the urban flood exposure index across 474 SA1s in the Maribyrnong and Moonee Valley LGAs. The index is derived from flood depth, elevation, and sand percentage in soil, capturing key physical determinants of flood susceptibility. The results reveal pronounced spatial heterogeneity in exposure across the study area. Approximately 11.60% of SA1s (55 units) exhibit very high exposure levels (index >0.037). The highest exposure is recorded in SA1 20,603,111,509 in Flemington (0.043), reflecting the combined influence of higher flood depth, lower elevation, and soil characteristics that increase inundation susceptibility. Other high-exposure SA1s are mainly clustered in Flemington and Ascot Vale, indicating a concentration of elevated exposure in the eastern part of Moonee Valley. A further 71 SA1s fall within the high exposure category,

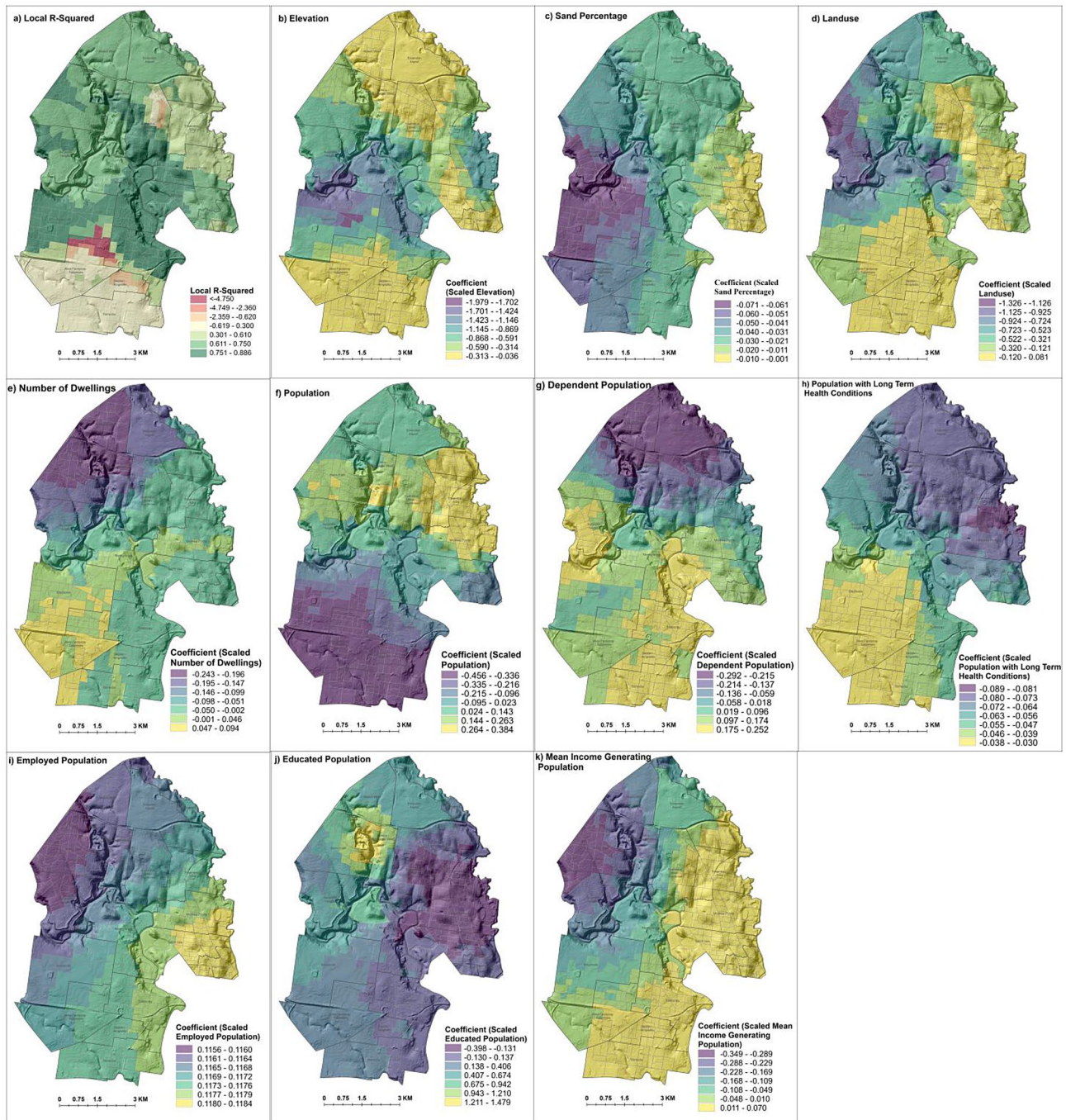


Fig. 4 (a) Spatial distribution of adjusted local R^2 of the MGWR model, (b–k) MGWR local coefficients for independent variables

reinforcing the clustering of vulnerable units along the northern and eastern corridors of the study area. Moderate exposure levels are observed in 62 SA1s, with values ranging from 0.033 to 0.035, mainly distributed across central transition zones.

Low to very low exposure levels are identified in 85 SA1s (index < 0.032). The lowest exposure is recorded in SA1 21,001,153,318 in Avondale Heights (0.004), associated with lower flood depth influence and more

favourable topographic conditions. These low-exposure SA1s are predominantly located in western and central areas, including Avondale Heights, Maribyrnong, and Braybrook, forming a coherent spatial cluster of reduced flood susceptibility. Figure 5(a) highlights a clear east–west gradient in exposure, with higher values concentrated in the eastern and northern corridors and progressively lower exposure toward the western part of the study area.

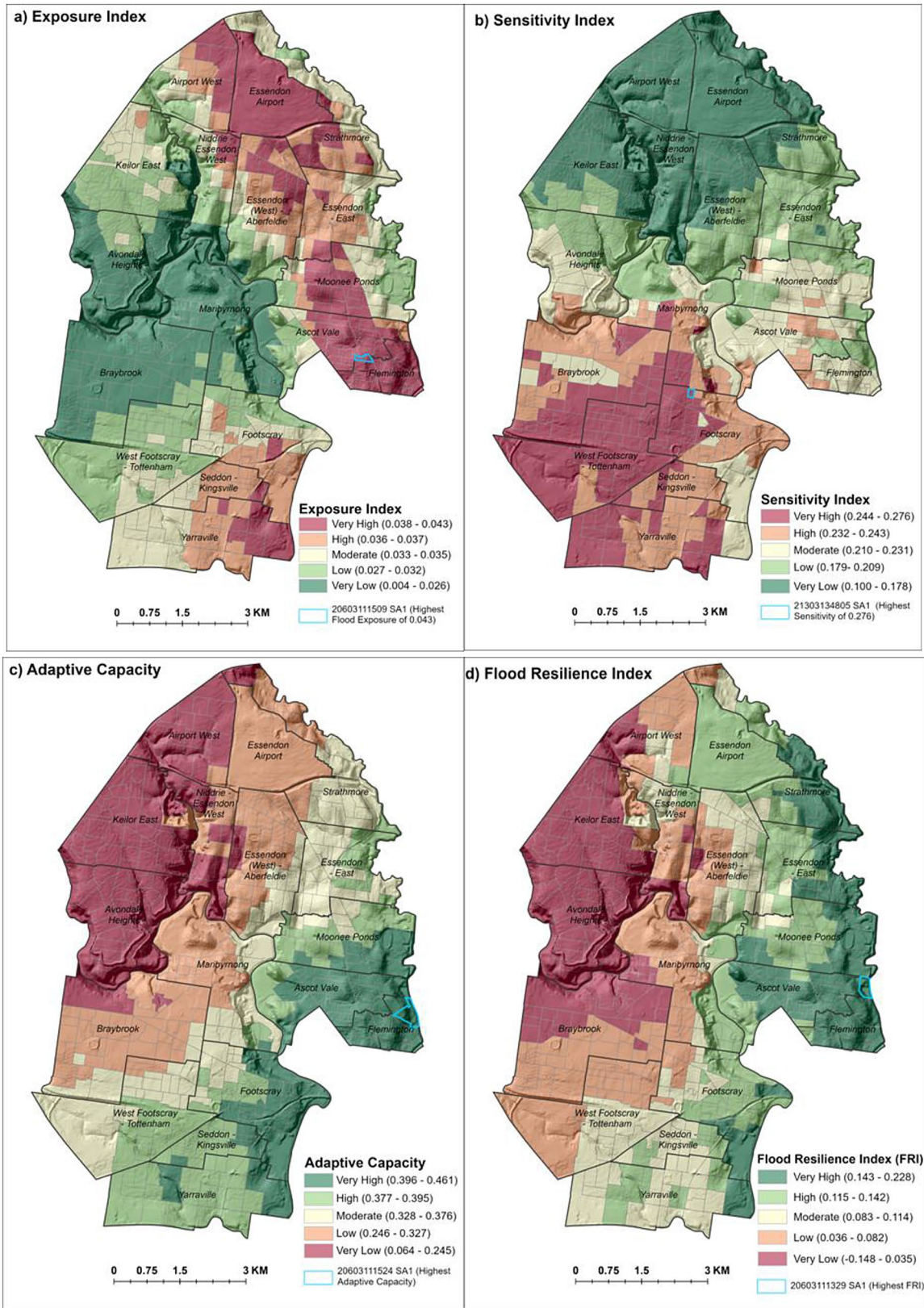


Fig. 5 Spatial variations of computed indices (a) Exposure Index, (b) Sensitivity Index, (c) Adaptive Capacity, and (d) Flood Resilience Index (FRI)

3.3.2 Urban flood sensitivity dimension

Figure 5(b) illustrates the spatial distribution of the urban flood sensitivity index across the 474 SA1s in the Maribyrnong and Moonee Valley LGAs. The index is derived from land use, number of dwellings, population, dependent population, and population with long-term health conditions, representing socio-economic vulnerability to flooding. The results show strong spatial heterogeneity in sensitivity across the study area. High and very high sensitivity values are predominantly concentrated along the southern corridor of the Maribyrnong River, particularly within Footscray, Maribyrnong, Braybrook, and West Footscray–Tottenham SA2s. The highest sensitivity is recorded in SA1 21,303,134,805 in Footscray (0.276), reflecting dense population structure and higher proportions of dependent and vulnerable groups. In total, 187 SA1s fall within the high to very high sensitivity classes, forming a continuous cluster across the southern part of the study area.

Moderate sensitivity is observed in approximately 20.46% of SA1s, distributed across eastern, western, and central transition zones. In contrast, low to very low sensitivity values are mainly concentrated in the northern parts of Moonee Valley, particularly around Airport West and adjacent SA2s, where socio-economic vulnerability is comparatively lower. The lowest sensitivity is recorded in SA1 21,001,122,609 in Airport West (0.100), indicating reduced population vulnerability and more stable socio-economic conditions.

3.3.3 Urban flood adaptive capacity dimension

Figure 5(c) presents the spatial distribution of the urban flood adaptive capacity index across the 474 SA1s in the study area. The index is based on employed population, educated population, and mean income-generating population, representing the ability of communities to respond to and recover from flood impacts. The results indicate strong spatial variability in adaptive capacity. High adaptive capacity is mainly concentrated in eastern SA2s, including Flemington, Ascot Vale, Moonee Ponds, and Footscray. These areas are characterised by stronger socio-economic conditions, including higher education levels, employment rates, and income capacity. The highest value is recorded in SA1 20,603,111,524 in Flemington (0.461), indicating strong recovery potential.

In contrast, low to very low adaptive capacity is primarily located in western and north-western SA1s, particularly Keilor East and Avondale Heights. A total of 186 SA1s fall within the low to very low categories, reflecting limited socio-economic resources and reduced capacity to recover from flood impacts. Moderate adaptive capacity is distributed across central transitional areas between high and low clusters.

3.3.4 Spatial distribution of flood resilience index

Figure 5(d) presents the spatial distribution of the Flood Resilience Index (FRI) across the 474 SA1s in the Maribyrnong and Moonee Valley LGAs. The FRI integrates flood exposure, sensitivity, and adaptive capacity to represent the overall resilience condition of urban areas under flood risk. The results show pronounced spatial heterogeneity in resilience across the study area, with values ranging from -0.148 to 0.228 . High and very high resilience values are predominantly concentrated in eastern SA2s, including Flemington, Ascot Vale, Moonee Ponds, Footscray, and Essendon Airport. These areas are characterised by a favourable combination of low exposure, reduced sensitivity, and strong adaptive capacity. The highest resilience is observed in SA1 20,603,111,329 in Ascot Vale, followed by multiple SA1s in Flemington, indicating a concentrated cluster of high-resilience urban units in the eastern corridor.

Moderate resilience is distributed across central transitional zones and parts of the southern study area, representing areas where exposure, sensitivity, and adaptive capacity are relatively balanced but not optimal. In contrast, low to very low resilience values are mainly concentrated in western and north-western SA2s, including Keilor East, Avondale Heights, Braybrook, Maribyrnong, and Airport West. The lowest resilience is recorded in SA1 21,001,153,434 in Keilor East (-0.148), reflecting the combined influence of high exposure, elevated sensitivity, and limited adaptive capacity. Figure 5(d) highlights a clear east–west gradient in flood resilience, with higher resilience concentrated in the eastern urban corridor and progressively lower resilience toward the western and north-western parts of the study area.

3.4 Damage assessment and Integrated Flood Resilience Index (IFRI)

Figure 6 presents the spatial distribution of the Flood Damage Index and the Integrated Flood Resilience Index (IFRI) across the 474 SA1s in the Maribyrnong and Moonee Valley LGAs. To complement the Flood Resilience Index (FRI), a Damage Index is developed to quantify potential socio-economic losses associated with flood exposure. This is subsequently integrated with the FRI to derive the IFRI, enabling a combined assessment of resilience capacity and realised flood impact.

The Damage Index shows strong spatial variability across the study area, ranging from 0 to 1. High and very high damage values are mainly concentrated in central and southern SA2s, including Ascot Vale, Footscray, Braybrook, Maribyrnong, and surrounding urban areas. The highest damage is recorded in SA1 20,603,111,307 in Ascot Vale (Fig. 6(a)), driven by high population density and a large concentration of exposed socio-economic assets. Overall, high-damage SA1s are dispersed across

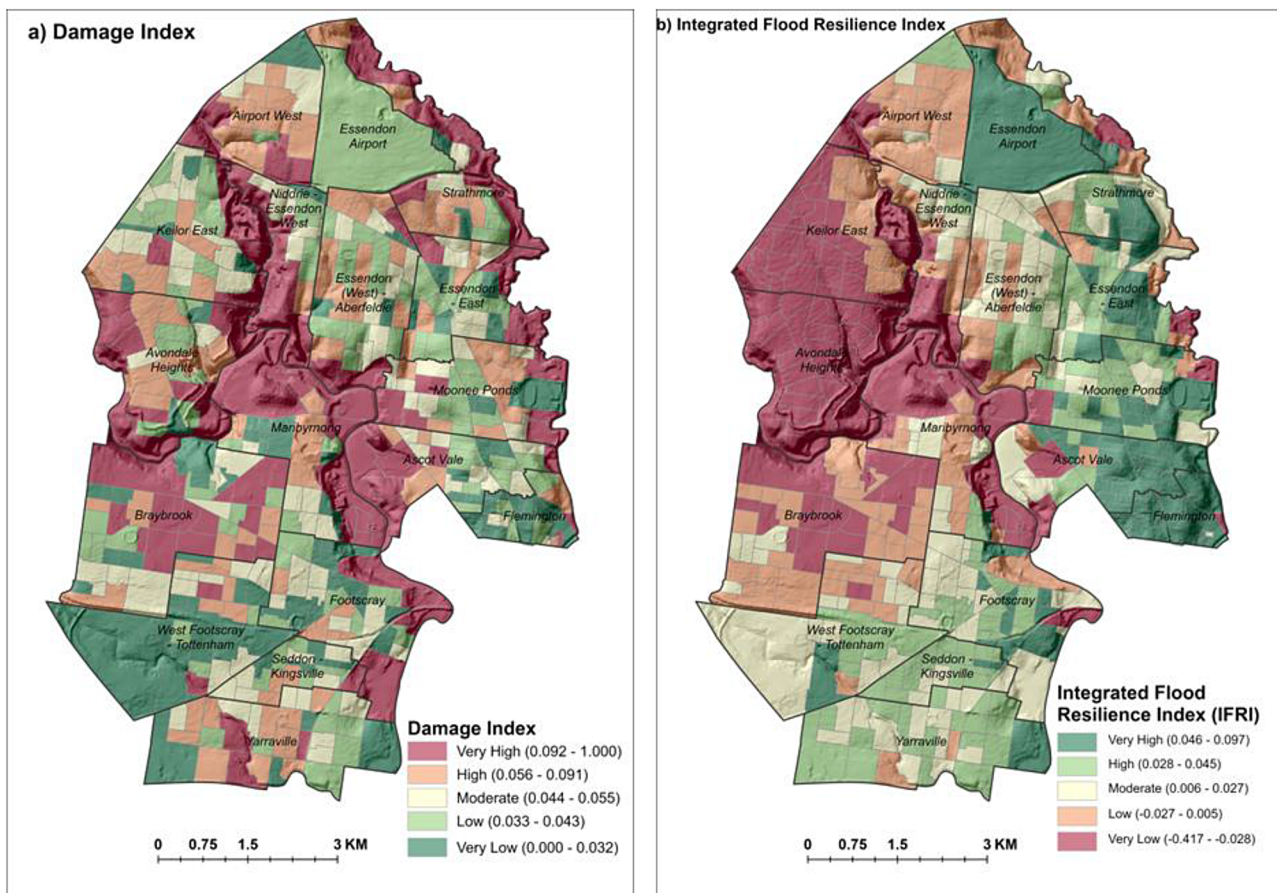


Fig. 6 Spatial variations of computed indices (a) Damage Index, and (b) Integrated Flood Resilience Index

multiple SA2s rather than forming a single continuous cluster, indicating widespread vulnerability among populated urban corridors. Low-damage areas are primarily located in western and northern SA1s, where population density and socio-economic exposure are comparatively limited. The lowest damage is observed in SA1 20,603,111,521 in Flemington, reflecting minimal population exposure and negligible potential for socio-economic loss. Figure 6(a) highlights a clear spatial contrast between high-impact urban cores and lower-impact peripheral zones.

The IFRI, shown in Fig. 6(b), integrates these damage patterns with exposure, sensitivity, and adaptive capacity to provide a comprehensive measure of flood resilience under realised impact conditions. High IFRI values are concentrated in eastern SA1s, particularly Flemington, Ascot Vale, Moonee Ponds, and Footscray, where lower exposure and sensitivity align with stronger adaptive capacity and reduced damage levels. Moderate IFRI values are distributed across central transition zones, reflecting mixed conditions where resilience and damage factors are relatively balanced. In contrast, low IFRI values are concentrated in western and north-western SA2s,

including Keilor East, Avondale Heights, Braybrook, and Maribyrnong. These areas are characterised by the combined influence of high exposure, elevated sensitivity, low adaptive capacity, and higher damage potential. The lowest IFRI is observed in SA1 20,603,111,307 in Ascot Vale, where extreme exposure and sensitivity coincide with maximum damage and limited adaptive capacity. Spatially, Fig. 6(b) reveals a consistent spatial imbalance in flood resilience, highlighting a clear east–west divide shaped by interacting environmental conditions and socio-economic disparities.

Figure 7 reveals that all the variables of availability indices are more or less positively correlated with each other, while IFRI is moderately correlated with the Damage Index.

3.5 Exceptional insights

For identified case studies in Maribyrnong and Moonee Valley LGAs, some key drivers appeared to have unexpected and counter-intuitive relationships to flood depth. In the northern catchment, an increase in dwellings was related to lower flood depths, potentially highlighting a lack of drainage infrastructure provided by the large

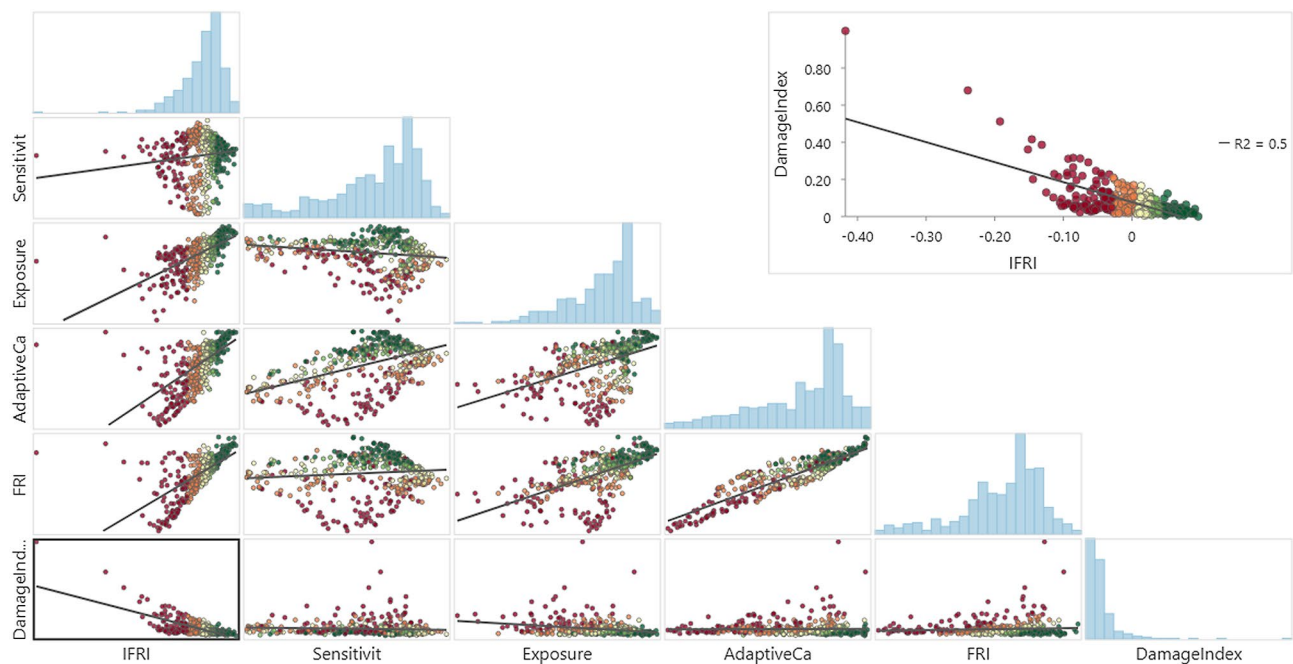


Fig. 7 Scatter matrix plot showing the Pearson's correlation with adjusted R2 for the constructed Exposure Index, Sensitivity Index, Adaptive Capacity, FRI, Damage Index and IFRI of 474 SA1s

grassland areas around Essendon Airport. Conversely, flood depths were lower in parts of the country with higher proportions of people with long-term health conditions (supporting the theory of inadequate community support infrastructure and outlier flood protection investments). These differences demonstrate the interaction between local conditions and infrastructure in the urban environment of flood vulnerability in cities.

Curiously, elevated levels of exposure index were identified in the northern and southeastern areas, such as Flemington, a finding that defied earlier predictions of increased exposure around rivers and creeks, where flood depth can be substantial. The index, based on flood depth, elevation, and sand content in the soil, indicated that, with low depths of flood, these areas appeared more vulnerable due to most extensive extents of elevation and greater percentage of sand. In addition, the weight scores these factors attributed to the index value could have heavily skewed the final value. Consequently, the aspects of these indicators that are weighted must be scientifically defensible and have a certain minimum amount of weight, as well as well-informed by experts and stakeholder input. This research indicates the spatial heterogeneity of flood resilience, and how the determinants vary from place to place.

Differences from the broader context of urban flood resilience are highlighted in the study in Maribyrnong and Moonee Valley LGAs because adaptive capacity, exposure, and sensitivity drive such deviations. Essendon Airport, near the Melbourne one in the graph, showed

high resilience to floods despite high exposure due partly to a highly educated and employed population, boosting adaptive capacity. Flemington showed high resilience however because of the effective management of the community and infrastructure despite only having moderate adaptive capacity. On the other hand, places like Footscray and Seddon-Kingsville had high sensitivity and adaptive capacity but only moderate resilience, emphasizing the curious dynamic of sensitivity, exposure, and adaptive capacity. Although Avondale Heights had minimal exposure, it had very little resilience largely because of its minimal ability to adapt to a changing situation and its high sensitivity to change, highlighting the extent to which education and employment drive resilience to flooding.

3.6 Policy implications

This study evaluates the resilience of 474 Statistical Areas Level 1 (SA1s) in Maribyrnong and Moonee Valley, Victoria, Australia, using the FRI, which integrates three dimensions: exposure, sensitivity, and adaptive capacity. The resilience of these SA1s was further refined using the IFRI, which accounts for potential flood damage. It finds that most metropolitan areas have low exposure with high adaptive capacity, leading to high overall resilience. 40.92% of urban units have high exposure, particularly in the northern and eastern part of the study area, due to inadequate employed and mean income-generating population and high rates of flood depth, and low elevation, and sand percentage in soil. 40.50% of urban units

are highly sensitive due to large populations and dwellings number, high illiteracy, and economic instability. Higher adaptive capacity is found in 40.50% of urban units, mainly in the eastern SA1s, due to the high number of educated and employed population. 41.35% of SA1s have a high damage index distributed all over the two LGAs due to a higher area ratio impacted by flood depth and a less financially stable population. The study uses the advanced spatial analysis method (MGWR) to assess local variations in vulnerability, revealing significant spatial patterns. Policymakers can use these insights to target interventions, improve infrastructure, enhance social safety nets, and promote economic growth in vulnerable areas.

SA1s identified as high-risk in Maribyrnong and Moonee Valley require targeted interventions, including upgraded drainage systems and levees, flood-resilient building practices, enhanced support for vulnerable populations, and strengthened emergency planning and early warning systems (Aitsi-Selmi et al., 2016). Moderate-risk areas would benefit from community education on flood preparedness, the implementation of green infrastructure such as permeable surfaces and urban wetlands, and improved coordination among local emergency services. Low-risk areas should focus on maintaining resilience through regular infrastructure maintenance, protection of natural floodplains, and continued support for economic and educational development. By integrating both physical flood exposure and socioeconomic vulnerability, the IFRI provides a comprehensive, people-centered framework that enables policymakers to guide zoning, allocate resources effectively, and enhance social capacity (Cutter et al., 2010), offering a more holistic approach than traditional strategies like the Victorian Floodplain Management Strategy.

4 Limitations and future work

Despite the comprehensive methodology employed in this study, several limitations should be acknowledged. The socio-economic indicators used to compute the resilience index may not fully capture the complexity of human–environment interactions and the multidimensional nature of flood resilience. Some important social and institutional factors, such as community networks, emergency preparedness, and governance capacity, were not available at the required spatial resolution, and there is potential bias in indicator selection and weighting.

The study relies mainly on historical flood data, which may not fully represent future flood scenarios under climate change and ongoing urban development. This may limit the framework's ability to capture emerging vulnerabilities and long-term resilience challenges. The damage assessment assumes a uniform population distribution within each statistical area and that most structures are

located at ground level. These simplifications may lead to over- or underestimation of flood impacts, particularly in areas with multi-story buildings, elevated structures, or uneven population densities. The absence of detailed building footprint, floor-level, and fine-scale population data further limits the precision of damage estimation.

The MGWR analysis is not intended as an external validation of the Integrated Flood Resilience Index (IFRI), but as a spatial explanatory approach to examine local variations in the relationships between resilience and its contributing factors. A full validation would require independent empirical outcomes, such as post-flood recovery rates, infrastructure recovery time, insurance claims, or historical disaster response performance, which were not consistently available at the required spatial resolution. Although MGWR effectively captures spatial heterogeneity, it does not fully represent the dynamic temporal changes in resilience and vulnerability, which are influenced by seasonal variation, policy interventions, and post-disaster recovery processes.

Future research should incorporate dynamic and real-time data, climate change projections, and more detailed building and population datasets to improve resilience and damage assessment accuracy. Strengthening indicator weighting through stakeholder and expert input, integrating additional social and institutional variables, and involving community stakeholders in validation and planning processes would further enhance the robustness and practical relevance of the framework.

5 Conclusion

This study developed a comprehensive framework for assessing flood resilience in the LGAs of Maribyrnong and Moonee Valley by integrating hydrological modeling, MGWR analysis, and socio-economic indicators. It addressed the research questions by identifying the essential components of a flood resilience framework, examining spatial interactions among exposure, sensitivity, and adaptive capacity, and mapping variations in resilience across 474 urban units. The integration of a Damage Index enhanced the analysis by revealing how socio-economic and physical factors jointly influence vulnerability. Findings indicated that low-lying and socio-economically disadvantaged areas exhibit lower resilience, while regions with stronger adaptive capacity demonstrate higher resilience. These outcomes provide valuable insights for policy and planning, supporting evidence-based floodplain management, infrastructure investment, and community preparedness strategies. The proposed Integrated Flood Resilience Index (IFRI) offers a novel, data-driven approach that contributes to improving urban flood resilience assessment and guiding future resilience planning efforts.

Appendix

Appendix 1: Diagnostic information for the GWR and MGWR model

Statistics	GWR	MGWR
R – Squared	0.7546	0.7669
Adjusted R-Squared	0.6838	0.7242
AICc	916.095	831.622
Sigma-Squared	0.316	0.2757
Sigma-Squared MLE	0.2454	0.2331
Effective Degree of Freedom	368.109	400.647
Optimal GWR Bandwidth: 62 (K nearest neighbours)		

Appendix 2: Standardization and normalization

Standardization involves transforming variables to have a mean of zero and a standard deviation of one, ensuring that variables are on the same scale. The resulting predicted (scaled flood depth) for flood depth and coefficient values for all scaled explanatory variables are standardized using the formula:

$$Z = \frac{(X - \mu)}{\sigma}$$

where:

Z is the standardized value,

X is the original value,

μ is the mean of the dataset, and.

σ is the standard deviation of the dataset.

Normalization scales variables to a common range (e.g., between 0 and 1), facilitating comparison and interpretation of their relative importance. Normalization is applied to standardized datasets.

It is calculated using the following formula:

$$X_{norm} = \frac{(X - X_{min})}{(X_{max} - X_{min})}$$

where:

X_{norm} is the normalized value,

X is the original value,

X_{min} is the minimum value in the dataset, and.

X_{max} is the maximum value in the dataset.

Appendix 3: Calculation of Exposure, Sensitivity, and Adaptive Capacity.

The weighted sum of normalized indicators within each dimension is computed by multiplication of normalized values of indicators with their respective weights to obtain composite scores for exposure, sensitivity, and adaptive capacity. These composite scores provide insights into the relative vulnerability of different regions within the study area. The general equation for calculating the dimension of the resilience index is given as follows:

$$Dimension = \sum_{n=1}^n (w_n * norm i_n)$$

where:

Dimension represents the overall score for a particular dimension of the resilience index (exposure, sensitivity, or adaptive capacity),

w_n is the weight assigned to the n-th indicator,

$norm i_n$ is the normalized value of the n-th indicator,

n is the total number of indicators considered for this dimension.

$$Exposure = w_1 * norm flood depth \\ + w_2 * norm elevation \\ + w_3 * norm sandper$$

where:

w_1 is the criteria weight for the flood depth indicator,

$norm flood depth$ is the normalized value for flood depth,

w_2 is the criteria weight for the elevation indicator,

$norm elevation$ is the normalized value for elevation,

w_3 is the criteria weight for sand percentage in the soil indicator.

$norm sandper$ is the normalized value for sand percentage in the soil indicator.

Similarly, Sensitivity and Adaptive Capacity are calculated.

$$Sensitivity = w_1 * norm landuse \\ + w_2 * norm number of dwellings \\ + w_3 * norm population \\ + w_4 * norm dependent population \\ + w_5 * norm population with long \\ term health conditions$$

where:

w_1 is the criteria weight for the landuse indicator,

$norm landuse$ is the normalized value for landuse,

w_2 is the criteria weight for the number of dwellings indicator,

$norm number of dwellings$ is the normalized value for number of dwellings,

w_3 is the criteria weight for population indicator,

$norm population$ is the normalized value for population indicator,

w_4 is the criteria weight for the dependent population indicator,

$norm dependent population$ is the normalized value for dependent population,

w_5 is the criteria weight for population with long term health conditions indicator,

$norm population with long term health conditions$ is the normalized value for population with long term health conditions indicator.

$$\begin{aligned} \text{Adaptive Capacity} = & w_1 * \text{norm employed population} \\ & + w_2 * \text{norm educated population} \\ & + w_3 * \text{norm mean income} \\ & \text{generating population} \end{aligned}$$

where:

w_1 is the criteria weight for the employed population indicator,

norm employed population is the normalized value for employed population,

w_2 is the criteria weight for the educated population indicator,

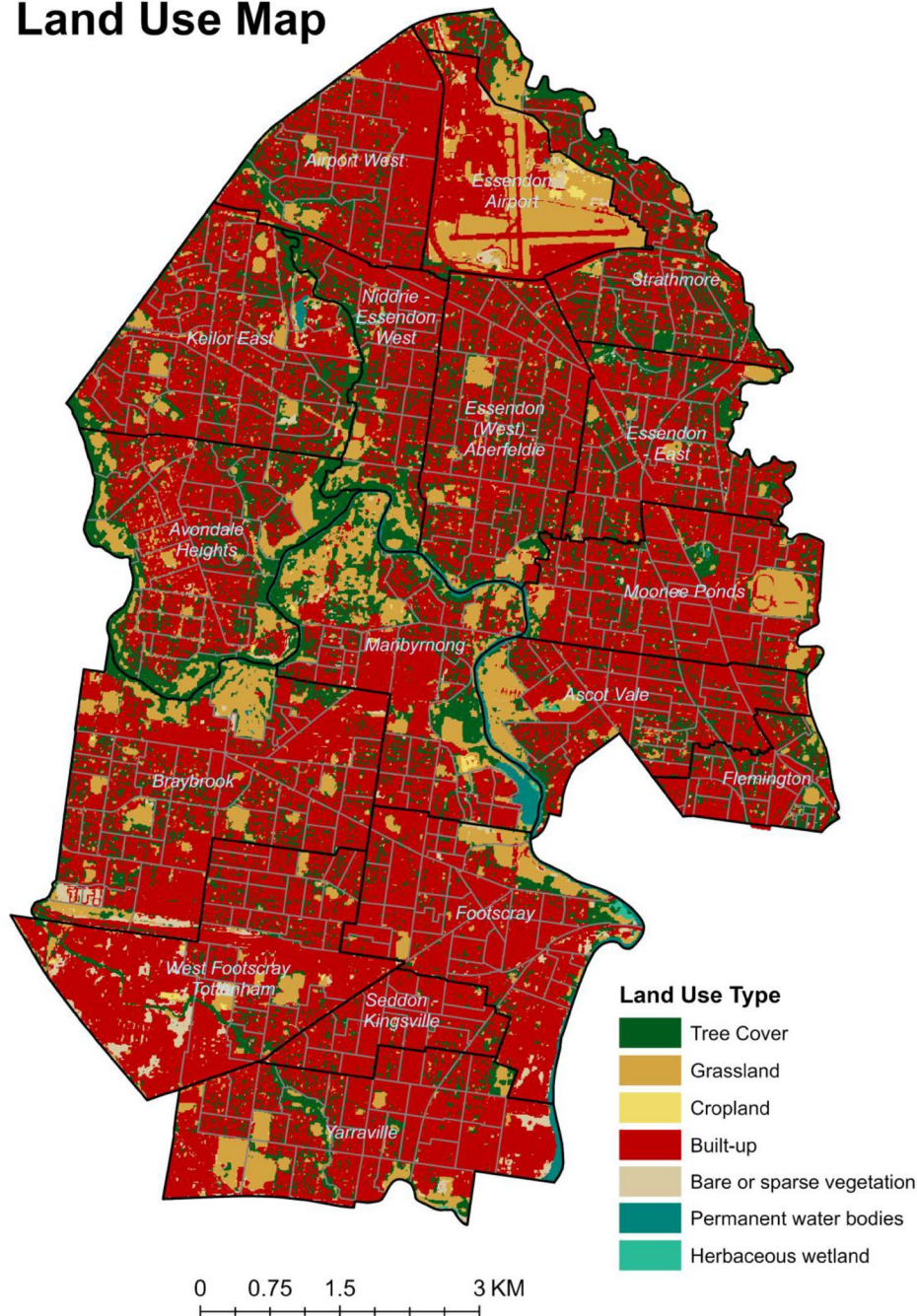
norm educated population is the normalized value for educated population,

w_3 is the criteria weight for mean income generating population indicator,

norm mean income generating population is the normalized value for mean income generating population indicator.

Appendix 4: Land use map of the study area

Land Use Map



Appendix 5: Datasets with their source link, format and authority

Description	Source Link	Format	Authority
Project Boundary	https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files#downloads-for-gda2020-digital-boundary-files	Esri Shapefile	Australian Bureau of Statistics (ABS)
Statistical Area Level 1 (SA1s) Boundary	https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files	Esri Shapefile	Australian Bureau of Statistics (ABS)
Statistical Area Level 2 (SA2s) Boundary	https://discover.data.vic.gov.au/dataset/vicmap-elevation-dem-10-m	Geo TIFF	Victorian State Government Department of Environment, Land, Water and Planning
Vicmap Elevation DEM 10 m	https://discover.data.vic.gov.au/dataset/vicmap-elevation-dem-10-m	Geo TIFF	Esri, ArcGIS Living Atlas of the World
Land use	https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=145.60457%2C-36.96071%2C7.392489711934156&mode=step&timeExtent=2017%2C2023&year=2023	Geo TIFF	Australian Bureau of Statistics (ABS)
MMA Administrative Boundary	https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/access-and-downloads/digital-boundary-files#downloads-for-gda2020-digital-boundary-files	Esri Shapefile	Australian Bureau of Statistics (ABS)
Socio-economic data (Population, Number of dwellings, Dependent population (age group <20 and > 59), Population with Long Term Health Conditions, Employed Population, Educated Population, Mean Income Generating Population)	https://tablebuilder.abs.gov.au/webapi/jsf/tableView/tableView.xhtml	Excel file	Table Builder, Australian Bureau of Statistics (ABS)
Soil Grids of Victorian- Sand Percentage	https://discover.data.vic.gov.au/dataset/soil-grids-of-victorian-soil-sand	Geo TIFF	Victorian State Government Department of Environment, Land, Water and Planning
Rainfall and River Flow Rate Data	https://www.melbournewater.com.au/water-and-environment/water-management/rainfall-and-river-levels#/reader/230106A	Excel file	Melbourne Water

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Author contributions

Lama: Methodology, Software, Data Curation, Visualization, Formal analysis, Writing - Original Draft, Writing - Review & Editing. Sun: Supervision, Conceptualization, Investigation, Resources, Project administration, Validation, Writing - Original Draft, Writing - Review & Editing.

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Data availability

The data that support the findings of this study are not publicly available but are available from the corresponding author upon reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

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